

Assignment 3

Pathway and Network Analysis of Aplastic Anemia and Healthy Donors

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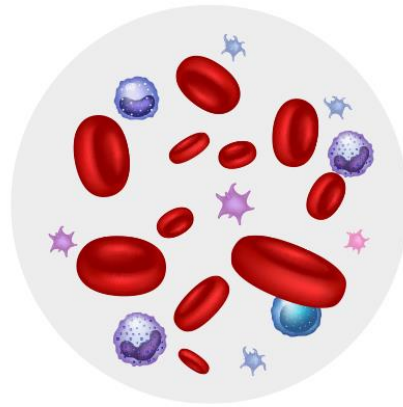
March 31, 2026

BCB420H1: Computational Systems Biology

Background



Normal



Aplastic Anaemia
(fewer red cells, white cells, and platelets)

- Rare and severe hematologic disorder
- Bone marrow failure and pancytopenia cause deficiency of all blood cell types

Zhu et al. (2019) investigated hematopoietic stem cells in patients with aplastic anemia (AA)

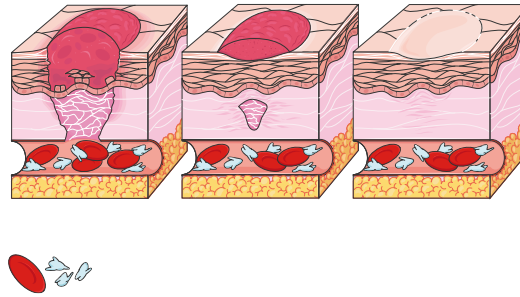
GSE165870: Bulk RNA-seq of AA & healthy donors

- Six non-severe (non-SAA) samples
- Three control donors
- 20234 genes (prior to quality control)

Motivations



Treatment limited
to management



Severe cases may
be life-threatening

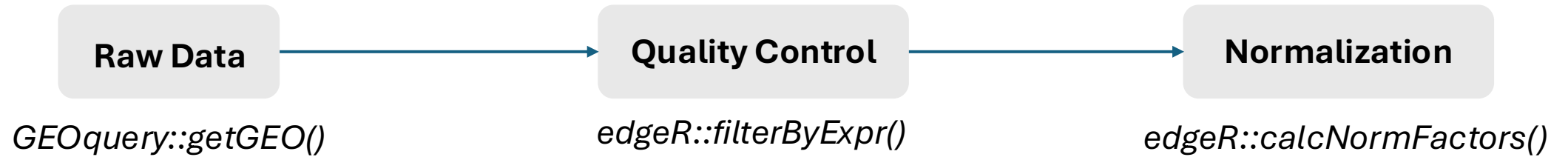


Existing studies
are observational

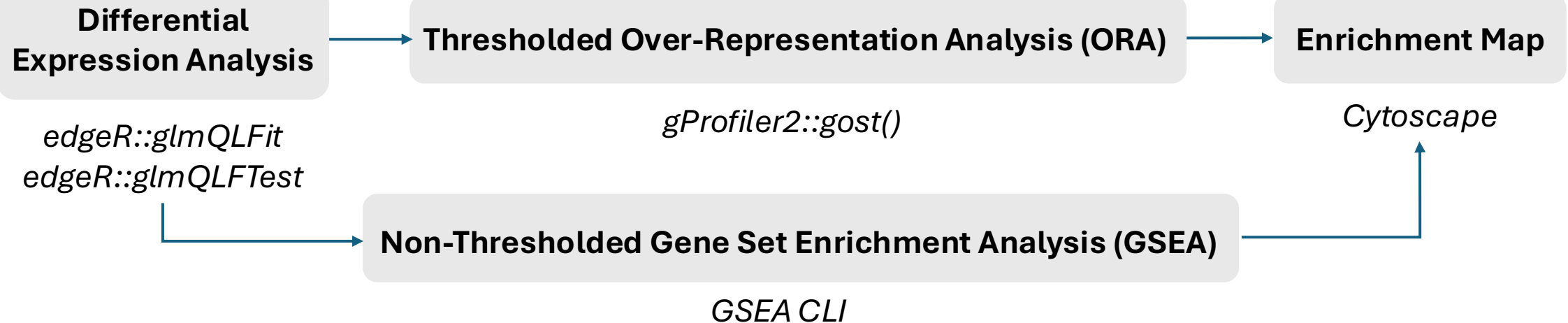
1. Is there a molecular signature stratifying severe and non-severe subtypes?
2. Is the single-cell transcriptomics results reflected in the bulk RNA-seq dataset?

Analysis Workflow

Preprocessing



Analysis

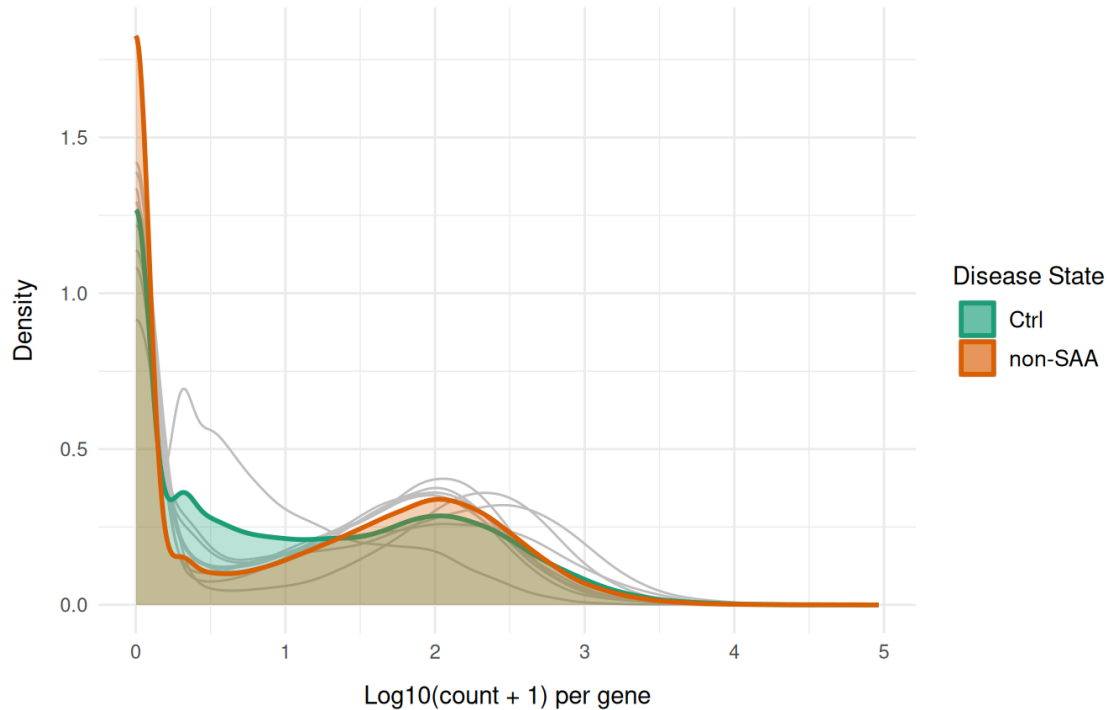


1. Introduction

Data Cleaning

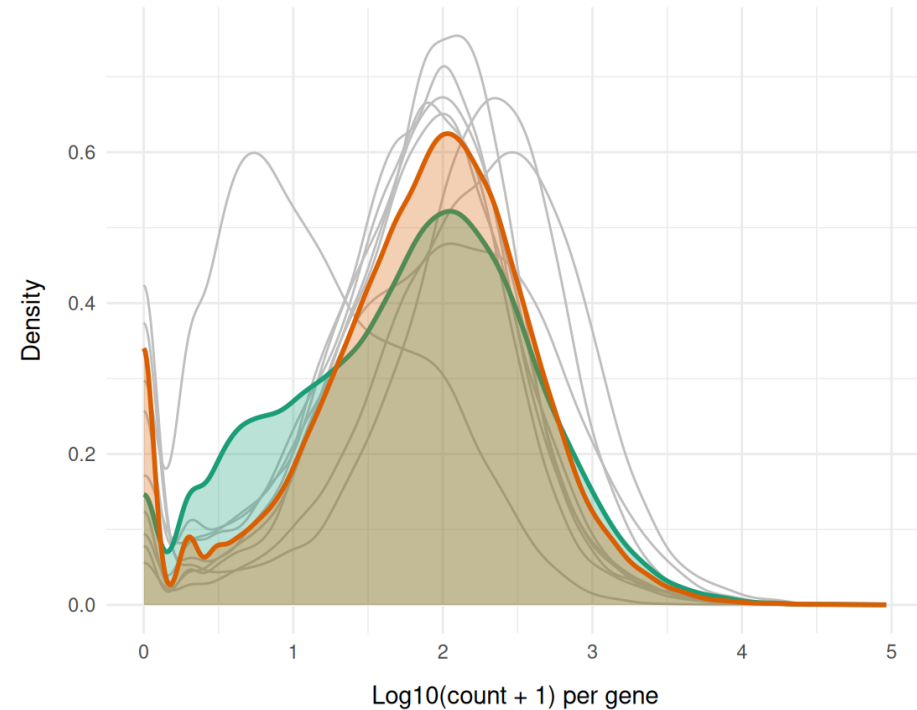
Distribution of Raw Gene Counts Across Samples

Grey lines: individual samples; coloured densities: pooled by disease state



Distribution of Gene Counts Across Samples

Grey lines: individual samples; coloured densities: pooled by disease state



Filtering Lowly Expressed Genes

`edgeR::filterByExpr()`

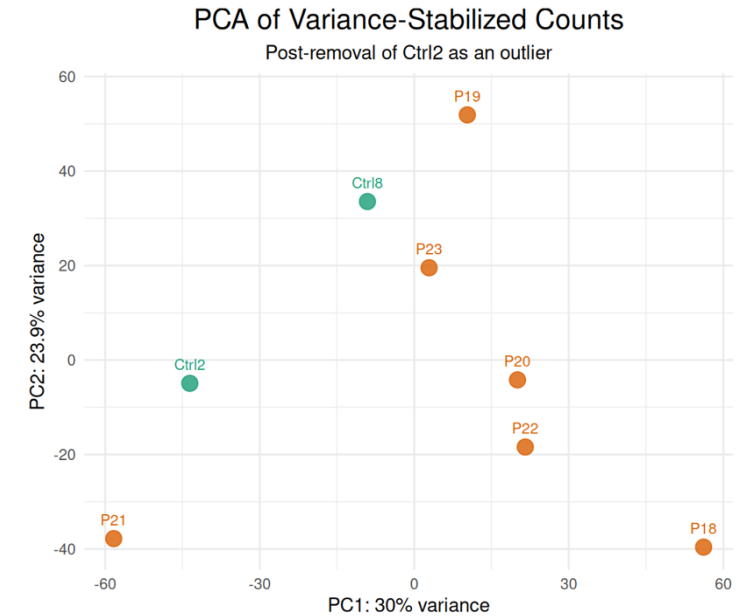
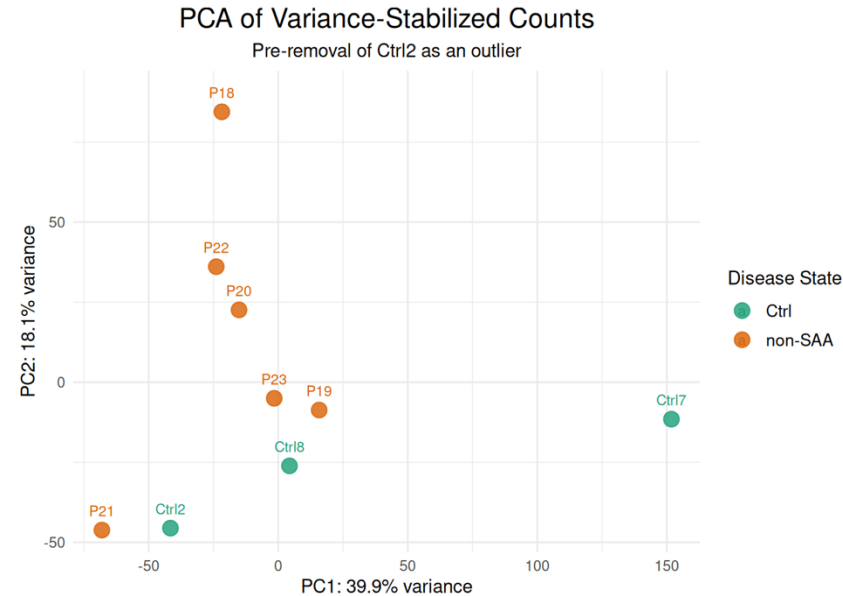
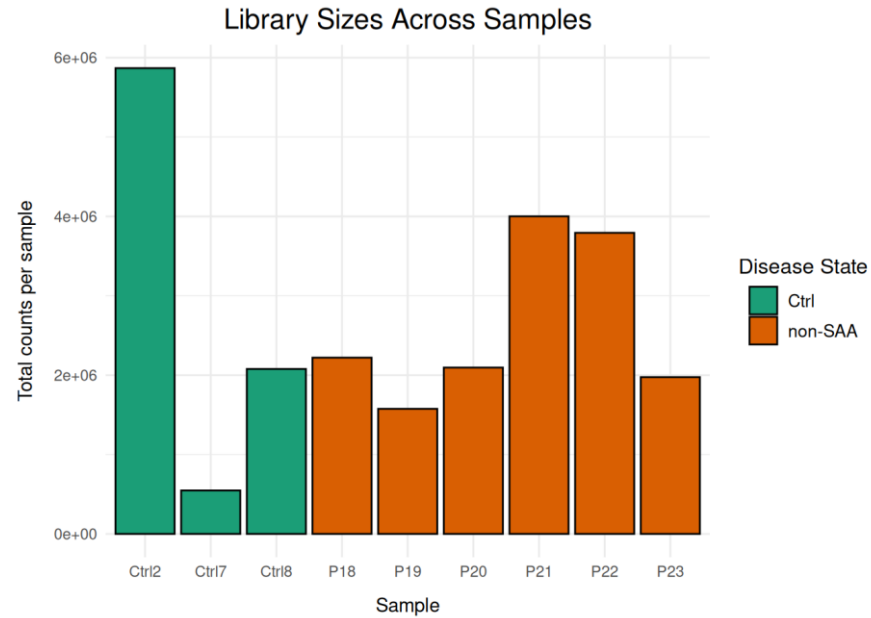
- Minimum three samples

54.3% Retention: 20234 → 10983

- Sufficient coverage for downstream

1. Introduction

Quality Control



Removing Technical Outlier

Directly removed from dataframe

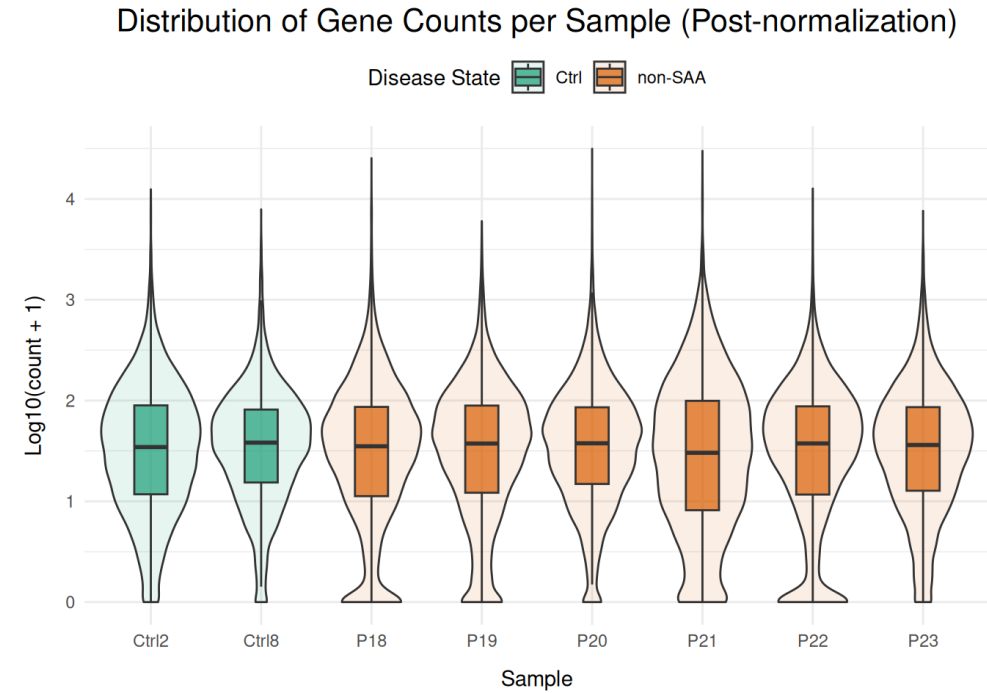
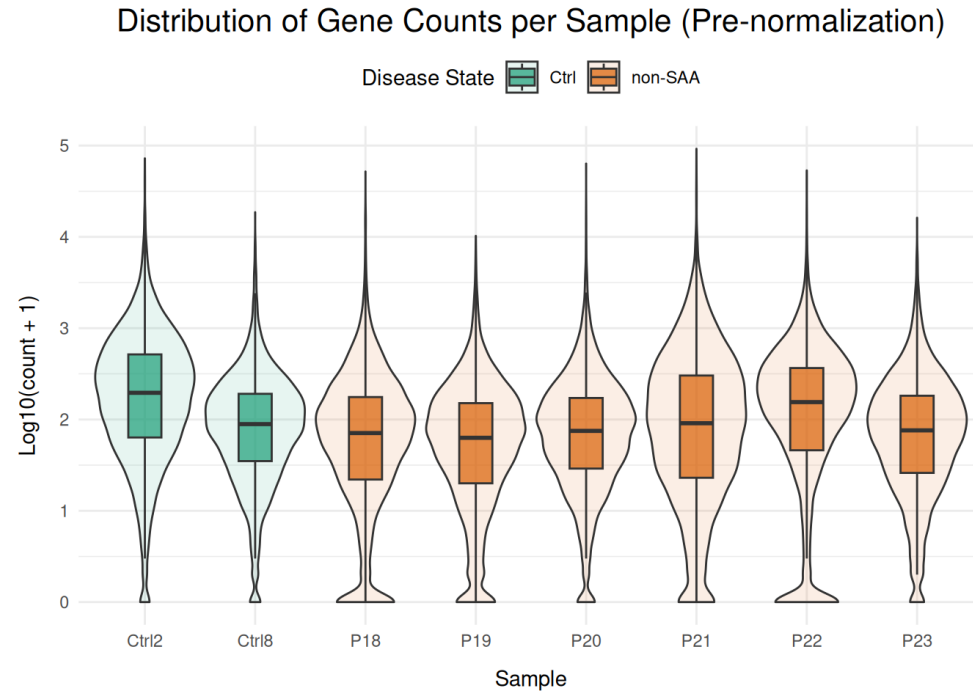
- *Ctrl7* extremely small library size
- Accounted for > 50% of variance

Post removal leaves eight samples

- Six non-SAA disease samples
- Two healthy controls

2. Methodology

Normalization

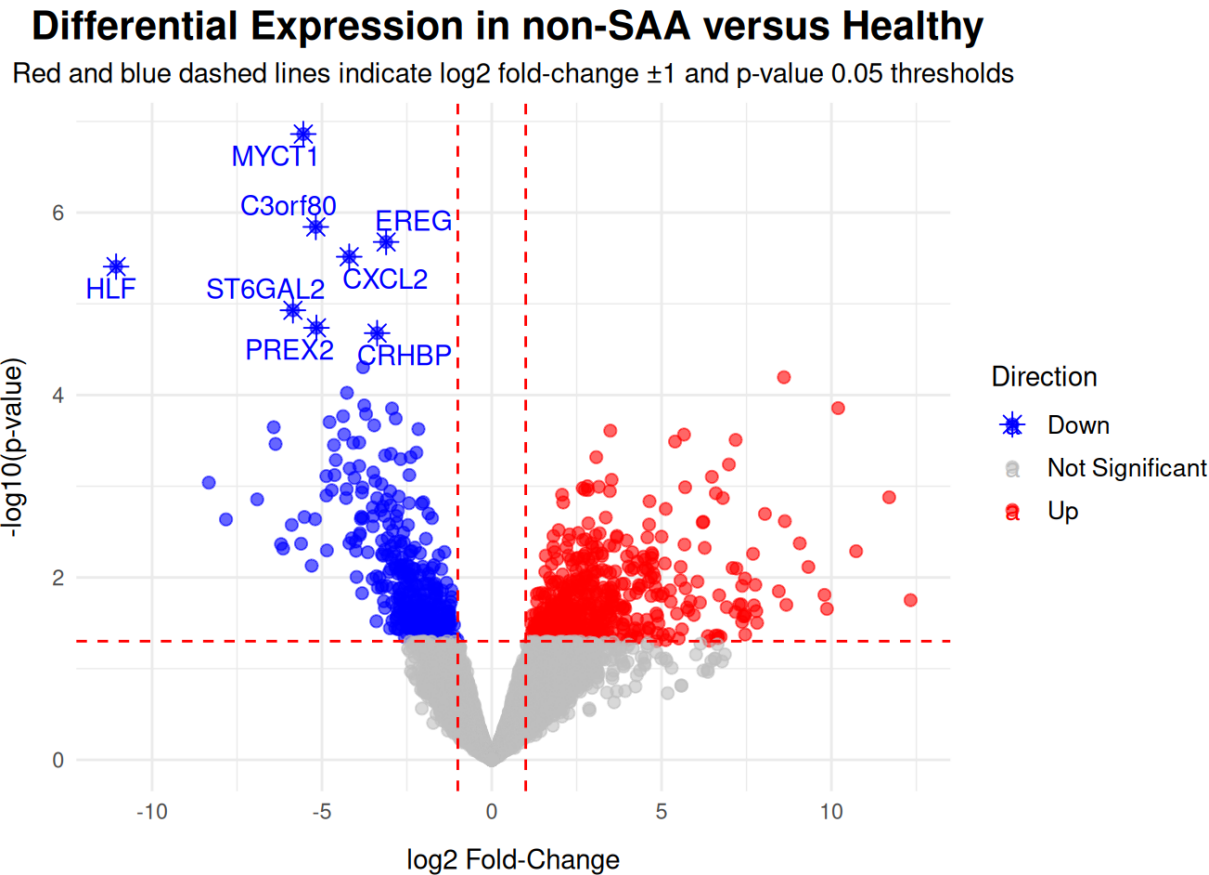


TMM Normalization

edgeR::calcNormFactors()

- Account for library composition
- Cross-sample comparisons
- Better for noisier datasets

Differential Expression Analysis



Differential Expression

`edgeR::glmQLFTest()`

- $P\text{-value} < 0.05$
- $(+) \log FC > 1.0$
- $(-) \log FC < -1.0$

n	Threshold
818	$P\text{-Value} < 0.05$
83	$FDR < 0.25$
8	$FDR < 0.05$

	gene_id	gene_symbol	logFC	FDR
MYCT1	ENSG00000120279.6	MYCT1	-5.549074	0.0015118
C3orf80	ENSG00000180044.5	C3orf80	-5.180615	0.0076881
EREG	ENSG00000124882.3	EREG	-3.110972	0.0076881
CXCL2	ENSG00000081041.8	CXCL2	-4.197923	0.0083700
HLF	ENSG00000108924.13	HLF	-11.060942	0.0085911
ST6GAL2	ENSG00000144057.15	ST6GAL2	-5.854468	0.0215475

Thresholded Over-Representation Analysis (ORA)

Enriched terms of Upregulated Geneset ($\log_2FC > 1.0$)

Source	Term Name	P-Value
GO:BP	Immune system process	0.0143209
GO:BP	B cell proliferation	0.0143209
GO:BP	B cell activation	0.0143209
GO:BP	B cell receptor signaling pathway	0.0143209
GO:BP	Immune response	0.0373322
GO:BP	Cell activation	0.0375036
GO:BP	Regulation of immune system process	0.0428806
GO:BP	Lymphocyte activation	0.0428806

Enriched terms of Downregulated Geneset ($\log_2FC < -1.0$)

Source	Term Name	P-Value
REAC	Cargo recognition for clathrin-mediated endocytosis	0.0225095
REAC	Clathrin-mediated endocytosis	0.0225095
REAC	EGFR interacts with phospholipase C-gamma	0.0225095
REAC	Inhibition of signaling by overexpressed EGFR	0.0225095
REAC	Signaling by overexpressed wild-type EGFR in Cancer	0.0225095
REAC	GRB2 events in EGFR signaling	0.0248508
REAC	SHC1 events in EGFR signaling	0.0255036
REAC	GAB1 signalosome	0.0402122

n	Category
24	Upregulated
59	Downregulated

gProfiler2::gost()

- *FDR* < 0.25
- *Min. size* = 5
- *Max. size* = 200

Annotation Databases: broad coverage & maintenance

- GO Biological Processes (2025-03-16)
- Reactome (2025-05-23)
- KEGG (2024-01-22)

Non-Thresholded Gene Set Enrichment Analysis (GSEA)

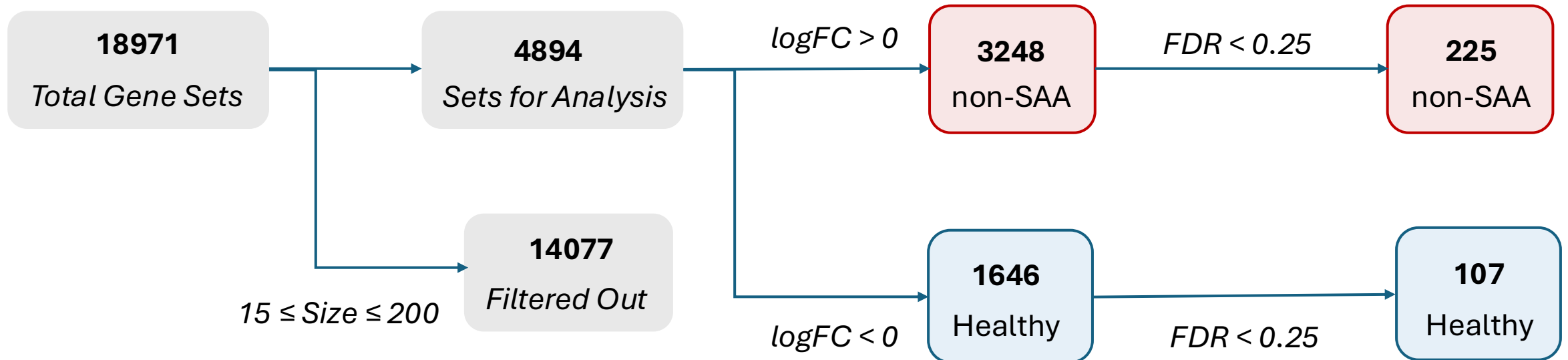
GSEA CLI (command-line)

- $FDR < 0.25$
- Min. size = 15
- Max. size = 200
- 1000 permutations
- Weighted scoring scheme

Gene Set Database: Bader Laboratory (September 1, 2025)

Ranking Score: negative log P-value * sign of logFC

(GSEA enrichment results on next slide)



3. Results

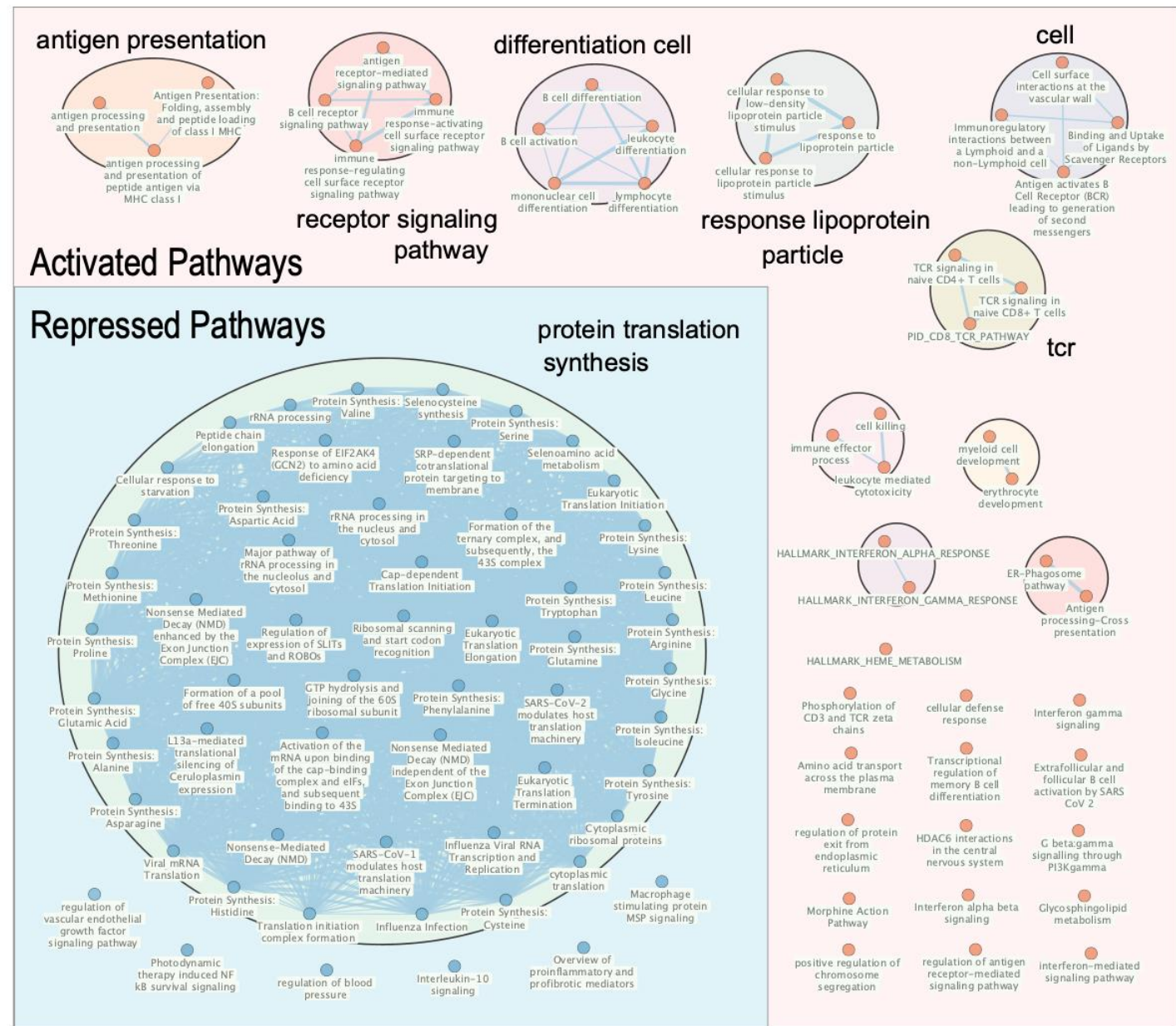
Enrichment Results

104 nodes and 1305 edges, 99.7% coefficient

n	Pathway Name
51	Protein translation & Synthesis
5	Cellular Differentiation
4	Antigen Presentation
4	Receptor Signaling
4	General Cell Processes
3	Lipoprotein Particle Response
3	T Cell Receptor Signaling

Cytoscape using AutoAnnotate

Phenotypes	AA (Up), Healthy (Down)
FDR Cutoff	0.05
Cluster Creation	clusterMaker2
Edge Similarity	Jaccard + Overlap
Edge Cutoff	0.1



3. Results

Major Themes

Non-SAA shows integrative upregulation of B- and T-cell activation and systemic suppression of EGFR signaling and endocytosis pathways

Coordinated **adaptive immune response**

- T-cell / B-cell activation
- Antigen presentation
- Receptor signaling

Systemic **metabolic slowdown**

- Inhibited protein translation
- Reduced cellular proliferation
- Dampened signaling (EGFR & clathrin-mediated)

Repressed

Overview of proinflammatory and profibrotic mediators

Photodynamic therapy induced NFkB survival signaling

regulation of vascular endothelial growth factor signaling pathway

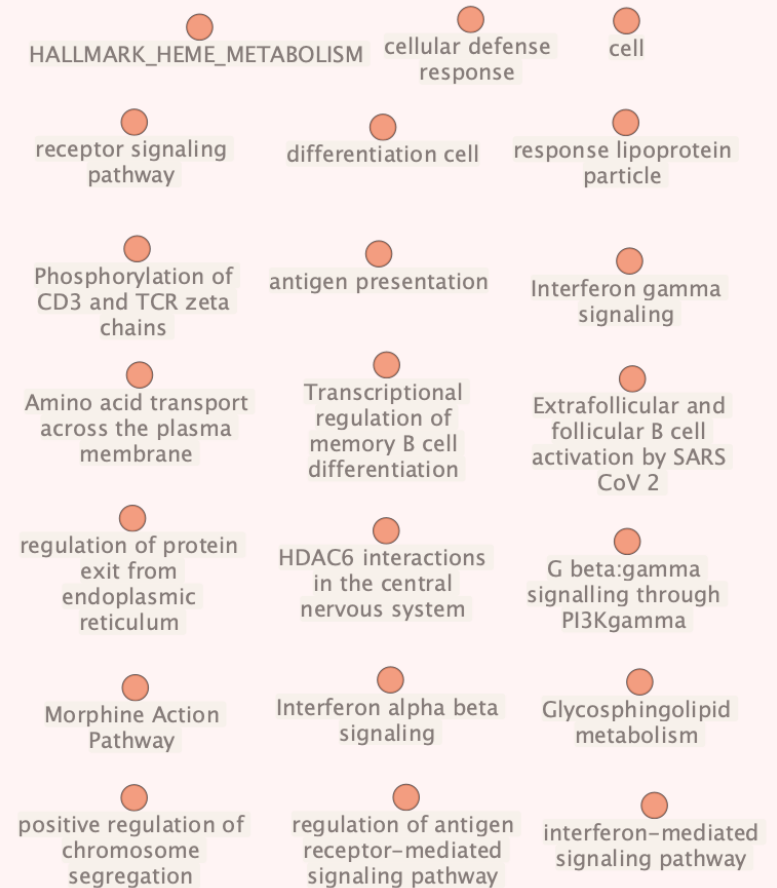
Macrophage stimulating protein MSP signaling

protein translation synthesis

Interleukin-10 signaling

regulation of blood pressure

Activated Pathways



Comparison of ORA versus GSEA Results

Largely consistent and complementary; overarching theme of adaptive immune activation

Direct comparison is not straightforward; rely on different statistical frameworks

ORA

Highlights Pathways Driven by Genes with Strongest Differential Expression

- Identified downregulation of EGFR signaling and clathrin-mediated endocytosis
- Reflects more pronounced changes in growth factor signaling & membrane trafficking

GSEA

Detects Coordinated Shifts across Many Genes within Pathways

- Identified downregulation of protein translation and synthesis
- Reflects coordinated decreases of ribosomal & translation-associated genes

Key Insights

Analysis agrees upon established immune-driven model and chronic inflammatory environment typical of non-severe aplastic anemia

Validated Immune Activation

- Upregulation of T-cell signaling, antigen presentation, and cytokine response agree with established model of immune-mediated stem cell destruction in AA

Corroborated Stress Response

- Broad suppression of translational machinery aligns with “stressed hematopoiesis” in literature
- Reflects downstream functional consequences of known splicing and polyadenylation defects

Identified novel broader signature of translational repression as a distinct pathological feature

5. Conclusion

References

R Packages

Chen, Yunshun, Lizhong Chen, Aaron T. L. Lun, Pedro Baldoni, and Gordon K. Smyth. 2025. “edgeR V4: Powerful Differential Analysis of Sequencing Data with Expanded Functionality and Improved Support for Small Counts and Larger Datasets.” *Nucleic Acids Research* 53 (2): gkaf018. <https://doi.org/10.1093/nar/gkaf018>.

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Annotation Databases

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Kanehisa, M., M. Furumichi, Y. Sato, M. Ishiguro-Watanabe, and M. Tanabe. 2021. “KEGG: Integrating Viruses and Cellular Organisms.” *Nucleic Acids Research* 49 (D1): D545–51. <https://doi.org/10.1093/nar/gkaa970>.

Package & Database Versions

edgeR (v. 4.8.2; 2025-12-23)

gProfiler2 (v. 0.2.4; 2025-11-10)

DESeq2 (v. 1.50.2; 2026-03-29)

Limma (v. 3.66.0; 2025-10-21)

GSEA (v. 4.4.0; 2026-01-30)

GO Biological Processes (R. 2025-03-16)

Reactome (R. 2025-05-23)

KEGG (R. 2024-01-22)

5. Conclusion

References

Literature Support

Bader Lab, University of Toronto. 2025. *Bader Lab Human Gene Sets, September 1, 2025 Release (Symbol Format)*.

https://download.baderlab.org/EM_Genesets/September_01_2025/Human/symbol/.

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Questions